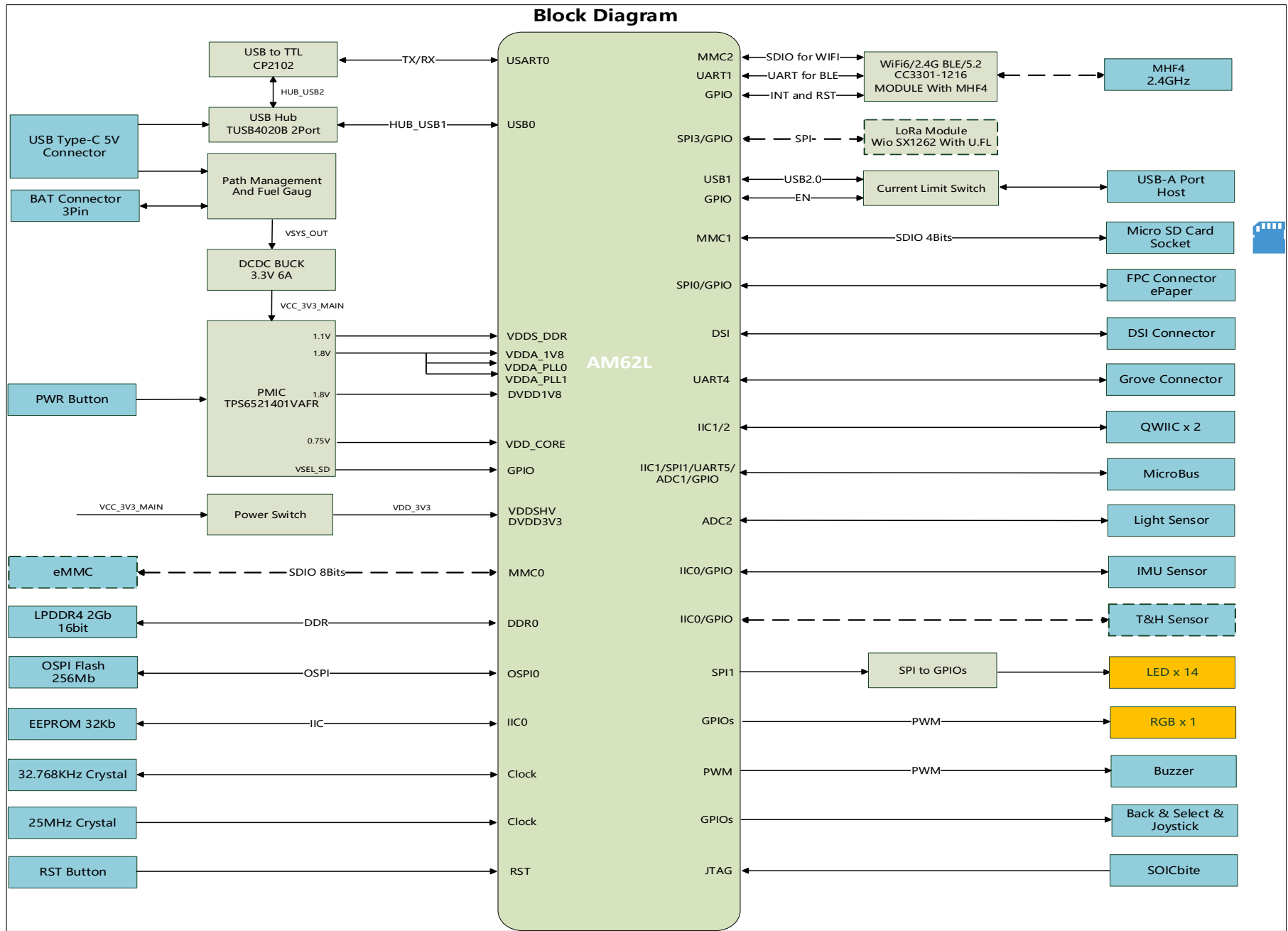


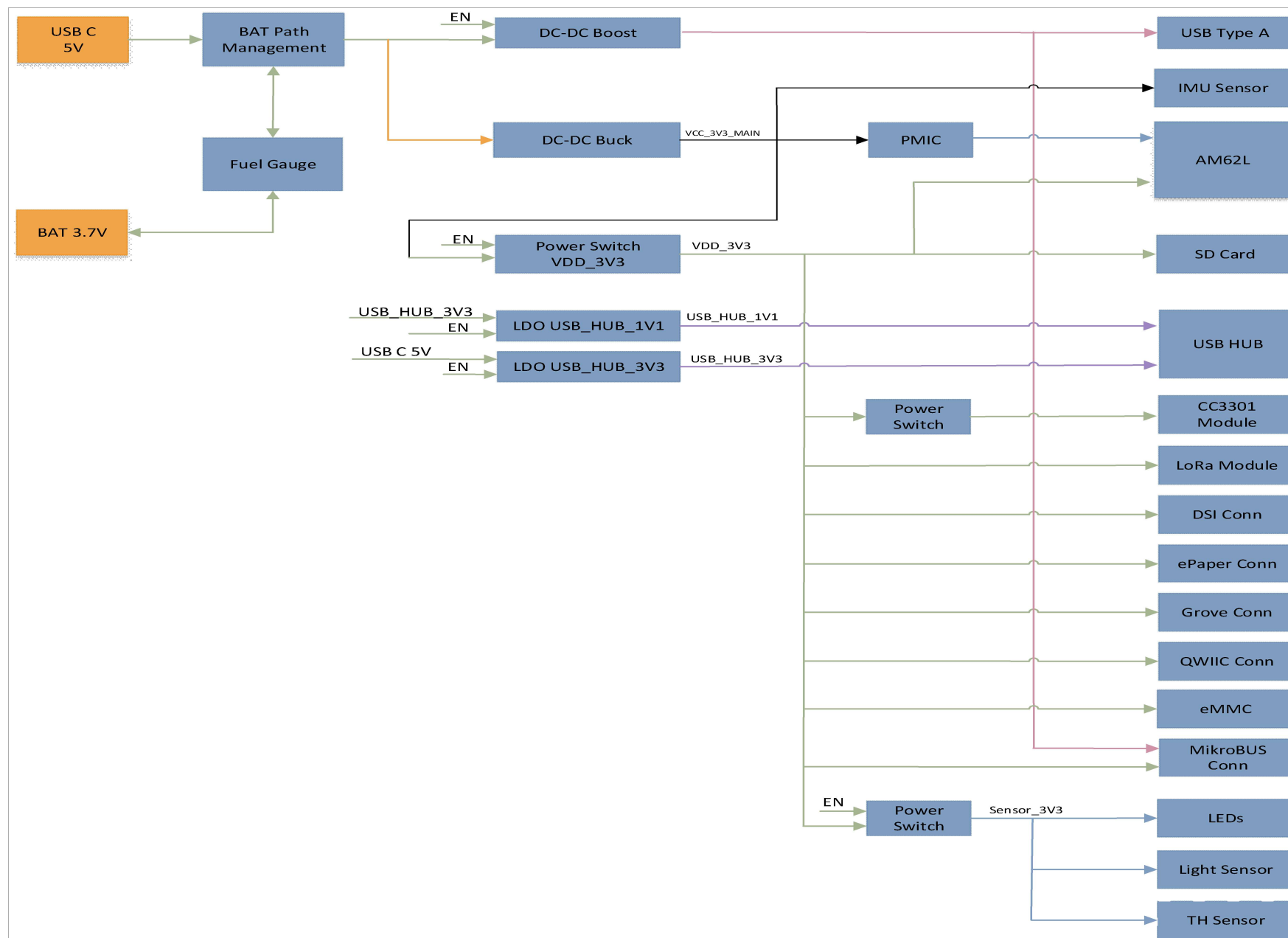
Schematic:Beaglebadge

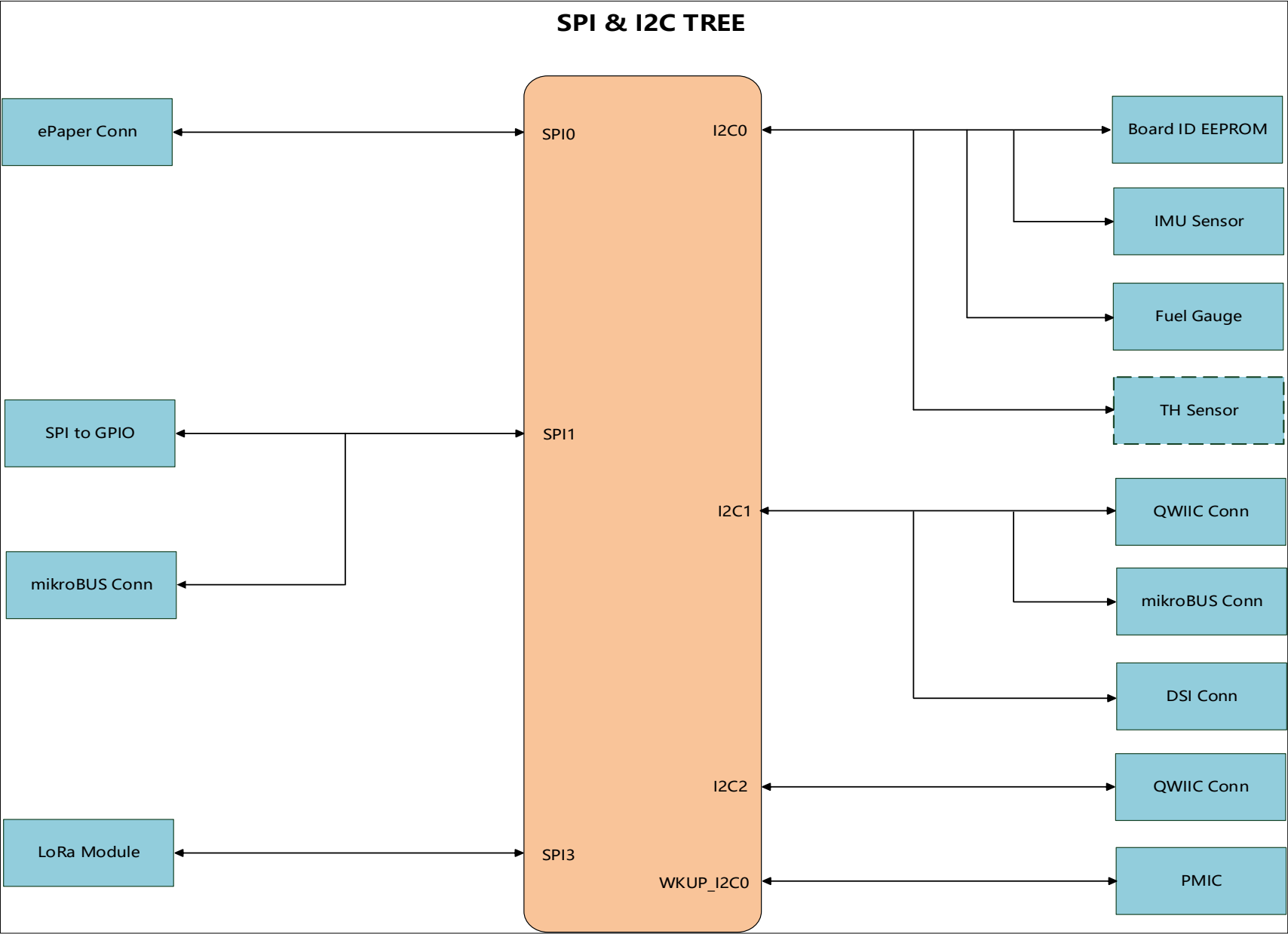
SHEET	SHEET NAME
01	Table of Contents
02	Block Diagram
03	Power Tree
04	IIC Tree
05	Power Input 5V
06	RTC ONLY Mode
07	AM62L PMIC
08	AM62L Power
09	AM62L DCAPs
10	AM62L WKUP & OSC & Reset
11	AM62L BOOT MODE
12	AM62L LPDDR4 DEVICE
13	AM62L OSPI Flash
14	AM62L Micro SD
15	AM62L ADC & DSI & OTHER
16	AM62L USB & HUB
17	USB2.0 Type A Host & UART
18	BM3301 & Sx1262 Module
19	EEPROM & SOICbite
20	Button & LEDs
21	Buzzer & IMU & Light
22	QWIIC & MikroBus
23	ePaper Conn
24	Other PINs & DSI & Grove
25	AM62L eMMC

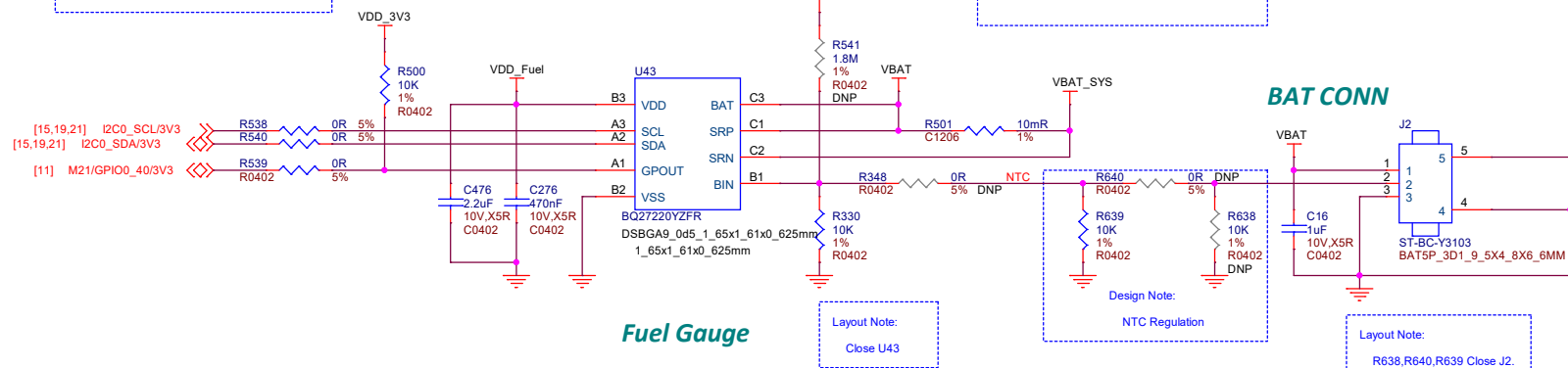
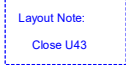
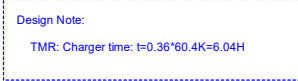
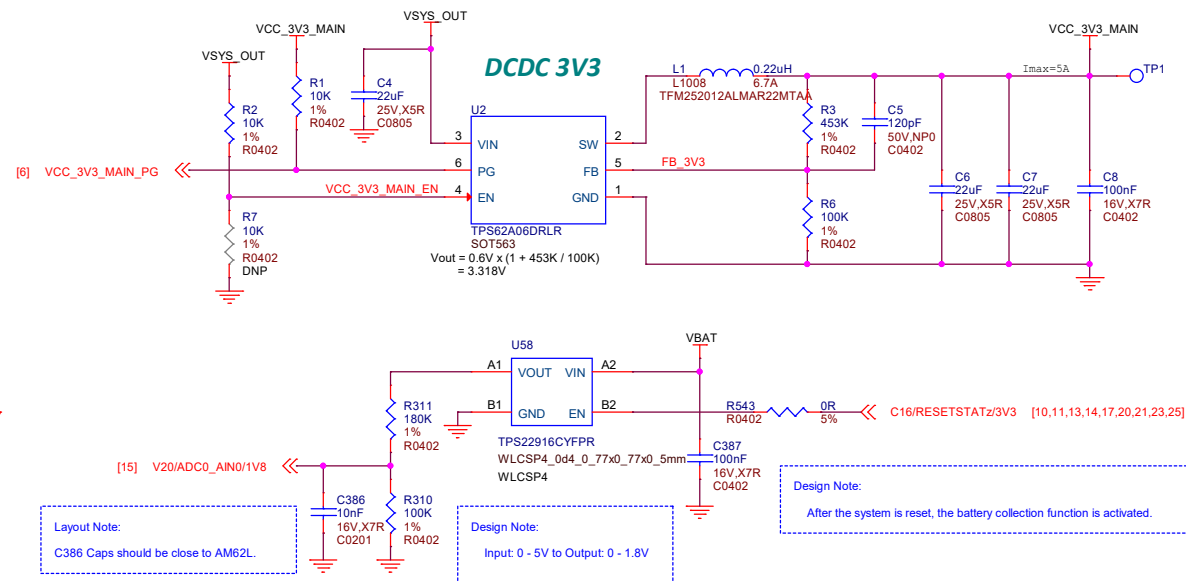
Revision History

DATE	REVISION	DESCRIPTION
2025/10/14	Rev A	1. Initial
2025/10/20	V0.3	Main modifications: 1. Change the capacitor withstand voltage to 2x or higher. 2. Correct the schematic symbol of the AND gate. 3. Add switch control to the power monitoring ADC. 4. Modify the circuit of RTC_PG and PMIC to reset AM62L. 5. DNI the output capacitor of TPS22965DSGR. 6. Add eMMC circuit. 7. Correct the wake-up signal(WAKEUP0/1). 8. Change Tag-Connect to SOIC16 Bite. 9. Change button gpios to 3V3 IOs because connect to SDIOBite 10. DNI I2C singles unpopulated for compatibility touchscreens.
2025/10/24	V0.4	1. Replace U9 with 0R resistors to switch Low Power mode. 2. Boot and Select function share one button. 3. Replace the Up/Down/Left/Right buttons with a joystick 4. Use the SPI1 to connect U42 (SPI to GPIO) and mikroBUS 5. Correct and add network identification descriptions 6. Adjust some GPIO allocation.
2025/10/28	V0.5	1. Added power supply selection for the IMU. 2. Modified the U62 circuit to prevent its input from floating. 3. Added the EXT_WAKEUP0 signal to pin 12 of J29 (SOICbite).
2025/10/29	V0.6	1. Use M23/GPIO0_43/3V3 to reset the USB Hub. 2. Use F22/GPIO0_29/PWM/3V3 to drive the buzzer. 3. The RGB LED has been changed to a smaller package.
2025/11/05	V0.7	1. Page8: Correct the VDDS0 power supply voltage to 1V8. 2. Page19: The JTAG signals change to 3.3V through a buffer. 3. Page24: The "Other GPIOs" pins shall be changed to 1V8.
2026/04/25	V1.0	1.Page5:R639 changed to NTC, charging cutoff at 45°C. R341= 2.4K, charging current reduced to 430mA. 2.Page6:Final power mode: RTC ONLY + IO + DDR. Removed main components U6, U8, U10, U39. 3.Page7:Added U88, reworked USB Hub power design. 3V3 and 1V8 only powered when Type-C is connected. 4. Page10:Removed U60, USB Hub now reset directly via GPIO. 5. Page18:Removed U36 LoRa module, added WiFi Power switch button. 6. Page20:Added Power button. D24 replaced with ESD0301L, added 220nF capacitor for ESD protection. 7. Page24:J27: removed AB9/RGMII2_RD0/1V8, added sensor 3V3. 8. Page25:DNP eMMC. 9.General:Minor passives adjustments across schematic.
2026/07/28	V1.0	1. Mount U36 as required. 2. PCB version number remains unchanged.

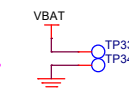








CE=H CHARGE STATE	STAT1	STAT2
Precharge	ON	ON
Fast charge	ON	OFF
Charge done	OFF	ON
Charge suspend	OFF	OFF



VDD_RTC_1V8, 0.5AMPS Supply

[5] VCC_3V3_MAIN_PG >> R27 10K R0402 0R 5% DNP

C27 1uF 10V X5R C0402

R25 10K 1% R0402

VCC_3V3_MAIN

VDD_RTC_1V8

R31 10K 1% R0402

RTCP_PG

U6 TPS74501PDRVR WSON6_0d65_2x2x0.8mm

IN 6

PG 5

OUT 1

EN 4

GND 7

FB 2

DNP

VDD_RTC_1V8_EN

C31 100nF 16V X7R C0402 DNP

R28 100K 1% R0402

R32 44.2K 1% R0402

VDD_RTC_1V8

C29 2.2uF 10V X5R C0402

TP23

Design Note:
For RTC ONLY MODE Used.

VDD_RTC (0.75V), 0.5AMPS Supply

Design Note:
For RTC ONLY MODE Used.

Design Note:
Low Power Mode: Output 0
PMIC Enable: Output 1

Design Note:
V_{IH}=1.39V

LOW POWER MODE SELECTION

PMIC_LPM_EN0/1V8

PMIC_EN [7,20]

PMIC_STBY [7]

RTC_PG

PORz_INPUT/1V8 [7,20]

VDD5_RTC_1V8

MIC_EN [7,20]

C35
100nF
16V X7R
C0402

U10

RTC_PG

R40
R0402

0R
5%

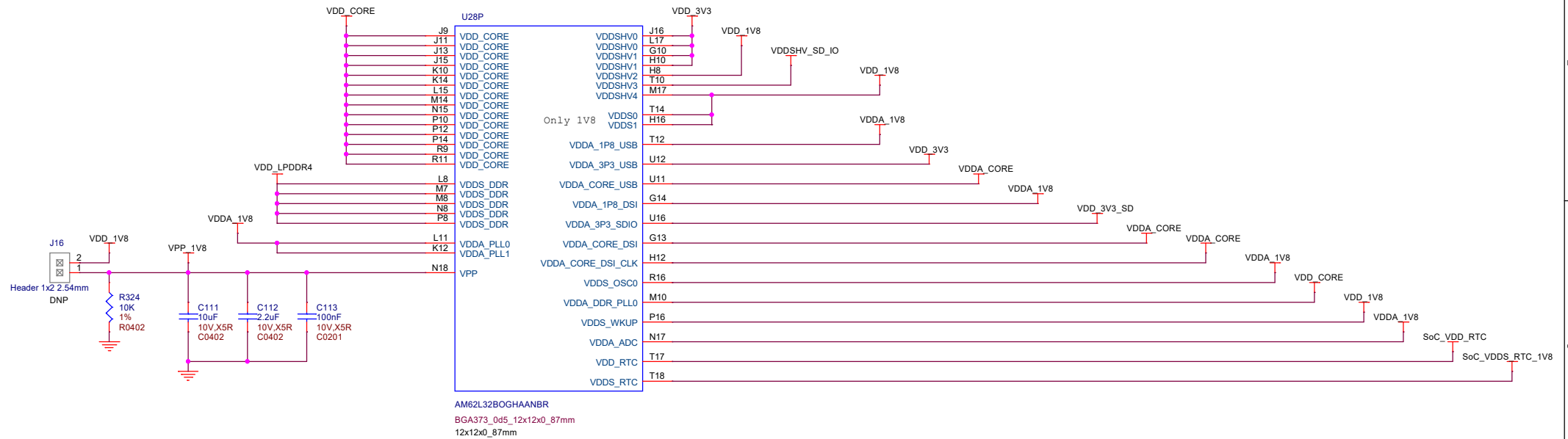
POR_z_INPUT/1V8 [7,20]

SN74LVC1G07DCKR
DCK5_2P15X1P4

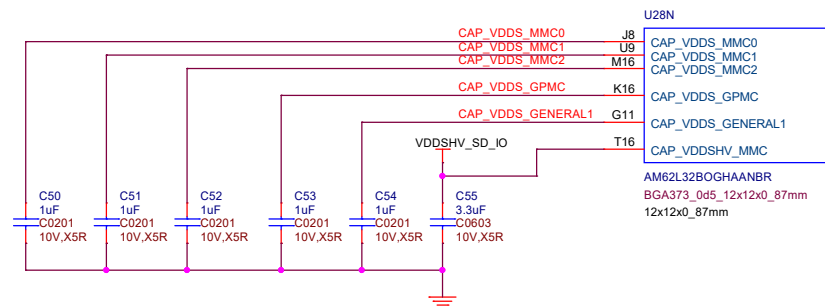
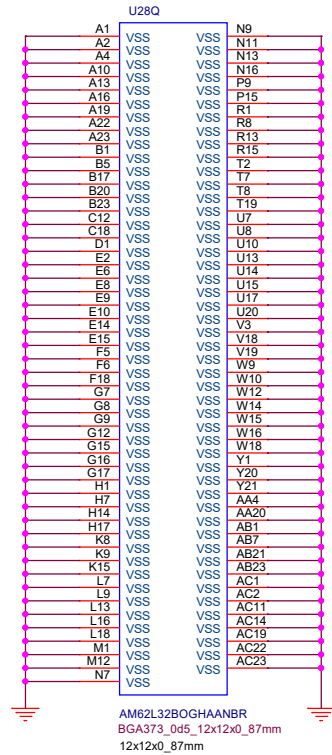
SC-70-5
DNP

RTC Only MODE	TO SOC	RTC + IO + DDR MODE
VDDS_RTC_1V8	SoC_VDDS_RTC_1V8	SOC_DVDD1V8
VDD_RTC	SoC_VDD_RTC	PMIC_VDD_RTC
RTC_PG	RTC_PORz	PMIC_GPO

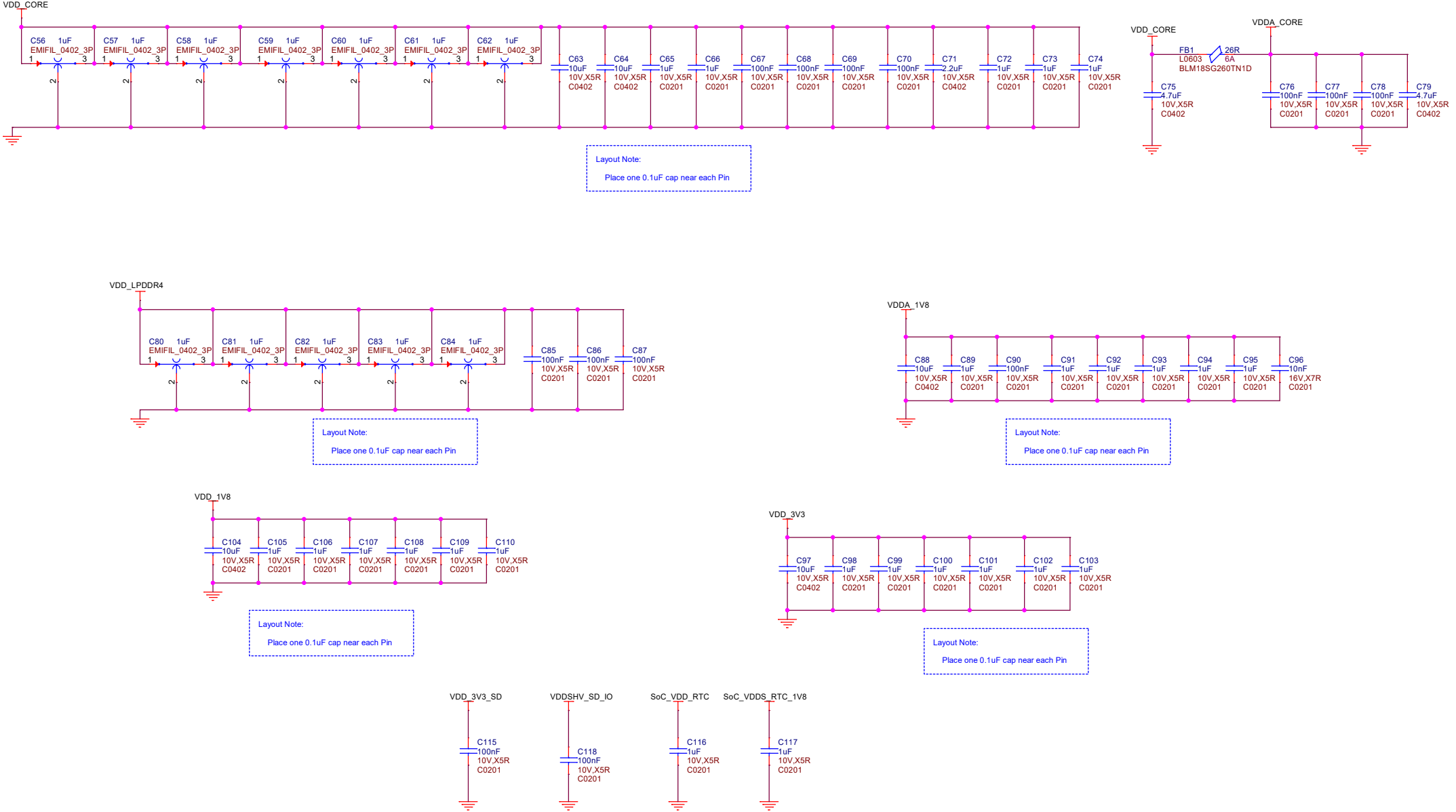
SOC POWER SUPPLIES AND SUPPLY RAILS



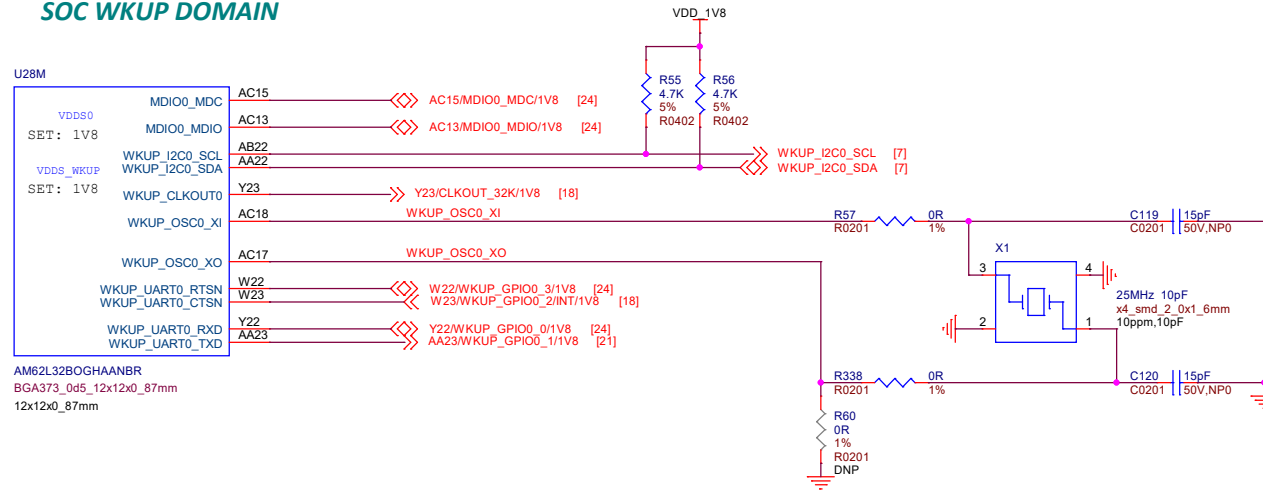
SOC VSS



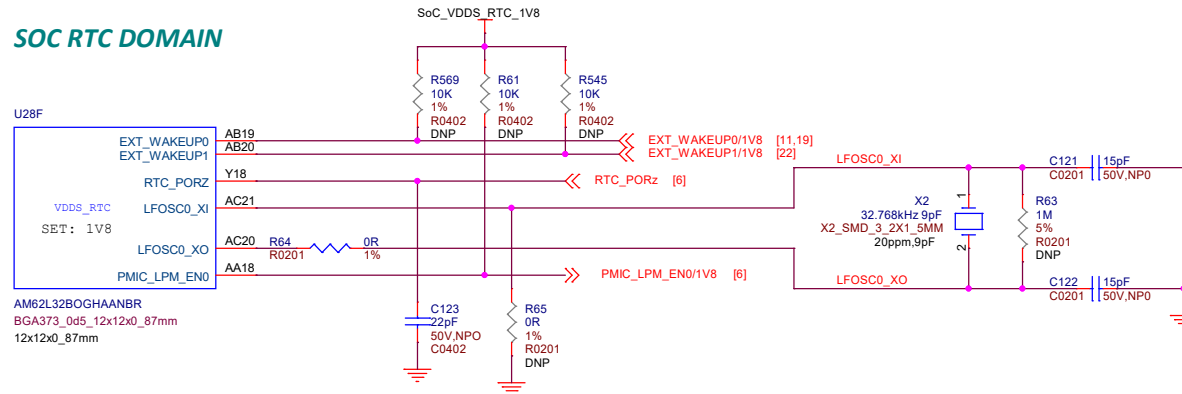
SOC POWER SUPPLIES - DECAPS



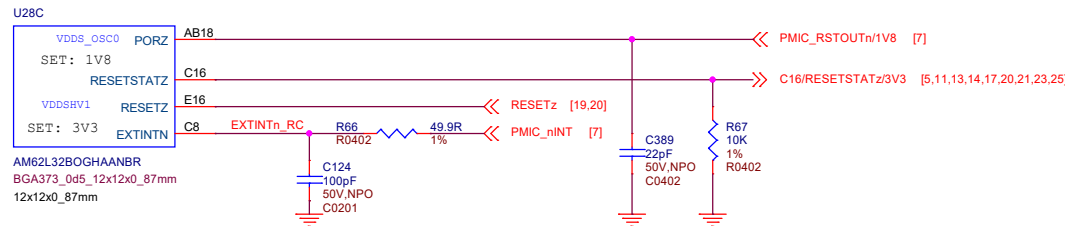
SOC WKUP DOMAIN

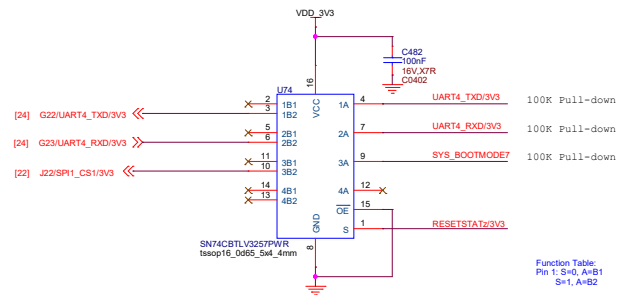


SOC RTC DOMAIN

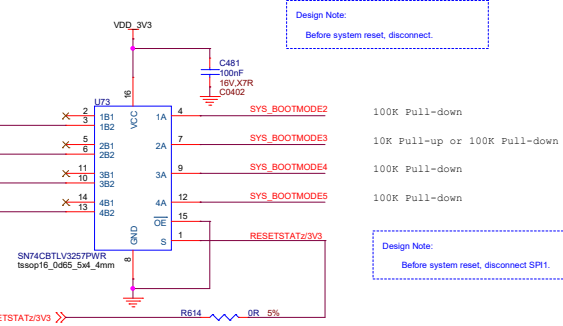


SOC RESET

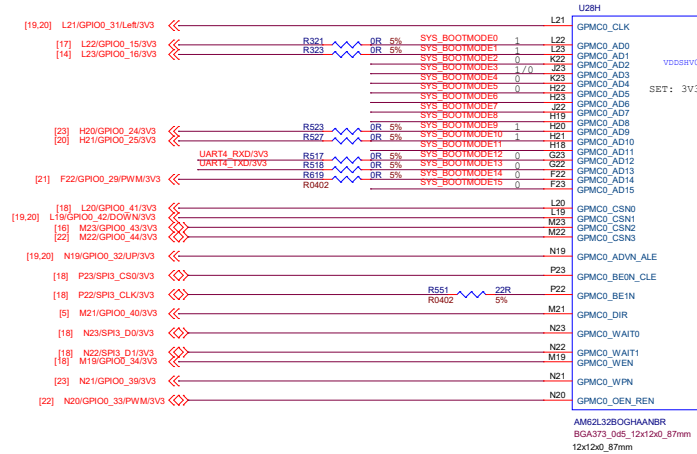




Function Table:
Pin 1: S=0, A=B1
S=1, A=B2

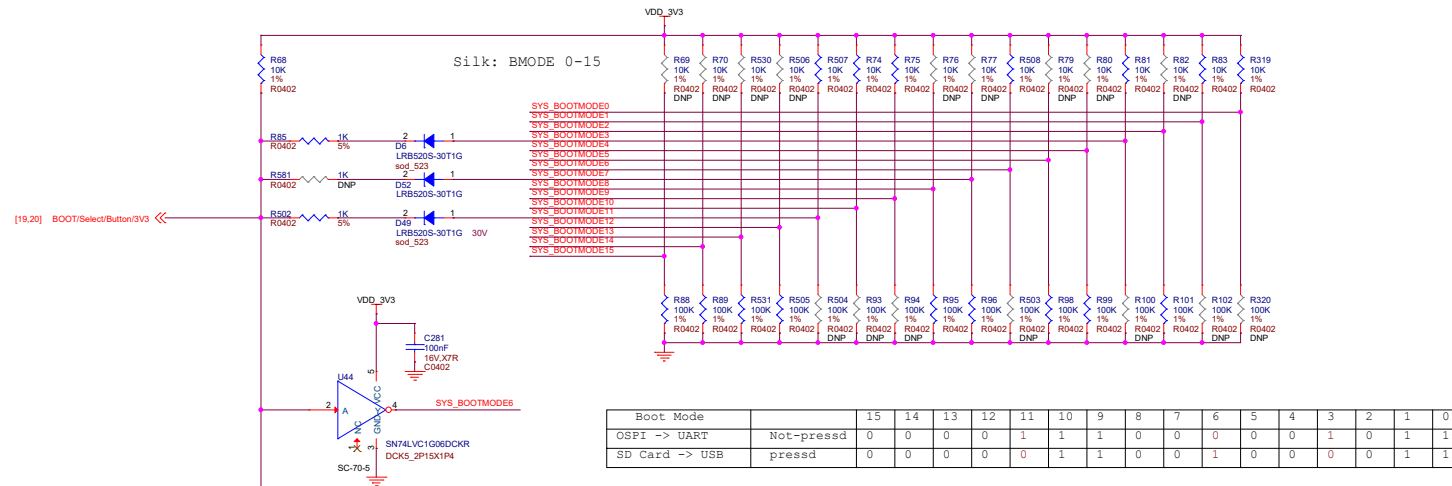


Design Note:
Before system reset, disconnect SPI1.



SOC - GPMC

SET: 3V3



Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OSPI -> UART	Not-pressd	0	0	0	0	1	1	1	0	0	0	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

Design Note:
Button Not-pressd:
1. PLL Config [B[2:0]] = 0b011 : Ref Colok -> 25MHz
2. Primary Boot [B[9:3]] = 0b1000001 : OSPI CS0 Boot
3. Backup Boot [B[13:10]] = 0b0011 : UART

Design Note:
eMMC BOOT:
1. BOOT7: (R77 and R581 Assembly, R96 to DNP)
2. BOOT6: (U44 to DNP)

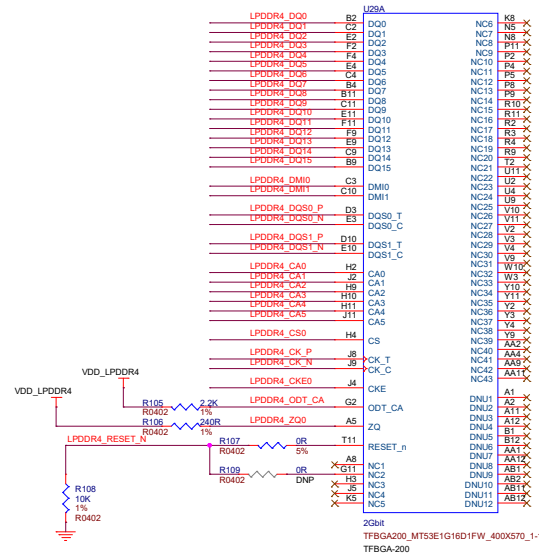
Design Note:
Button Not-pressd:
1. PLL Config [B[2:0]] = 0b011 : Ref Colok -> 25MHz
2. Primary Boot [B[9:3]] = 0b1011001 : eMMC Boot
3. Backup Boot [B[13:10]] = 0b0011 : UART

Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
eMMC -> UART	Not-pressd	0	0	0	0	1	1	1	0	1	1	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

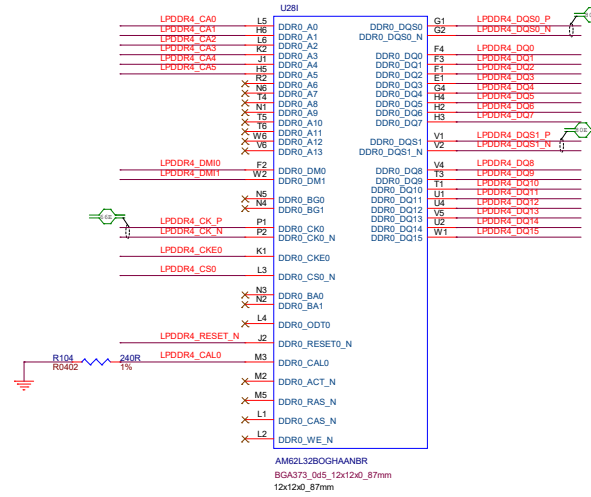
Design Note:
Use the Joystick_Press Button or Select Button to wake-up System.

LPDDR4 DEVICE

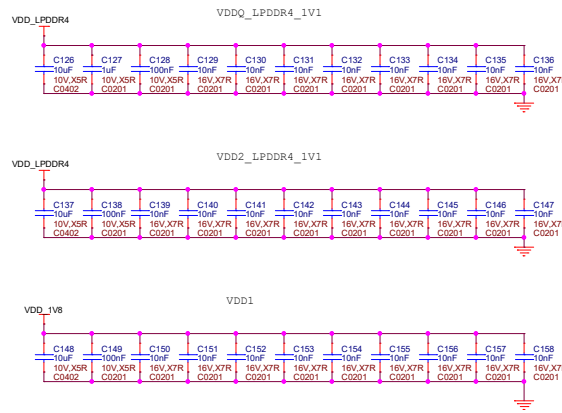
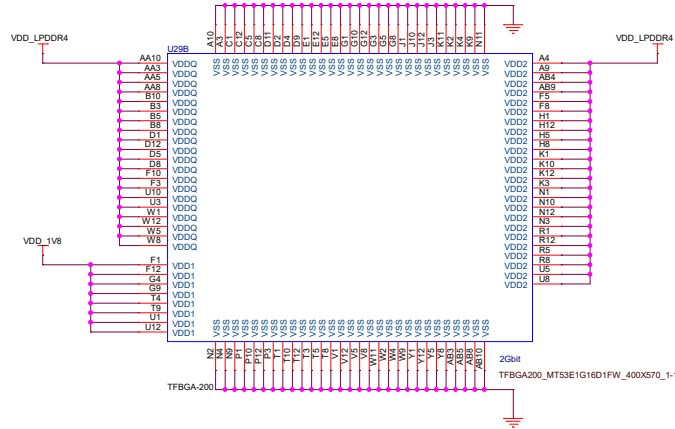
Silk: LPDDR4



SOC LPDDR4 INTERFACE



LPDDR4 POWER DECAPS



U28K

OSPI0_CLK

OSPI0_D0

OSPI0_D1

OSPI0_D2

OSPI0_D3

OSPI0_D4

OSPI0_D5

OSPI0_D6

OSPI0_D7

VDD_S1

1V8

OSPI0_CS0

OSPI0_CS1

OSPI0_CS2

OSPI0_CS3

OSPI0_DQS

OSPI0_LBCLKO

R552

R0201

OSPI_NOR_CLK

1%

C22

D21

E23

F21

F19

G20

F20

OSPI_DQ0

OSPI_DQ1

OSPI_DQ2

OSPI_DQ3

OSPI_DQ5

OSPI_DQ6

OSPI_DQ7

C20

D20

D18

C23

E22

E18

OSPI_NOR_CS

GPIO_NOR_RSTn

OSPI_NOR_INTn

OSPI_NOR_DQS

C23/GPI00_14/1V8 [18]

E18/GPI00_1/1V8 [25]

AM62L32BOGHAANBR

BGA373_0d5_12x12x0_87mm

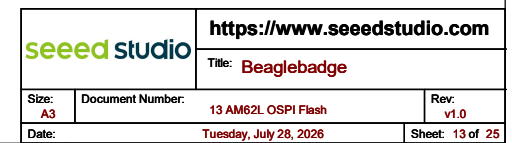
12x12x0_87mm

R623

1K

5%

R0402

[illegible]

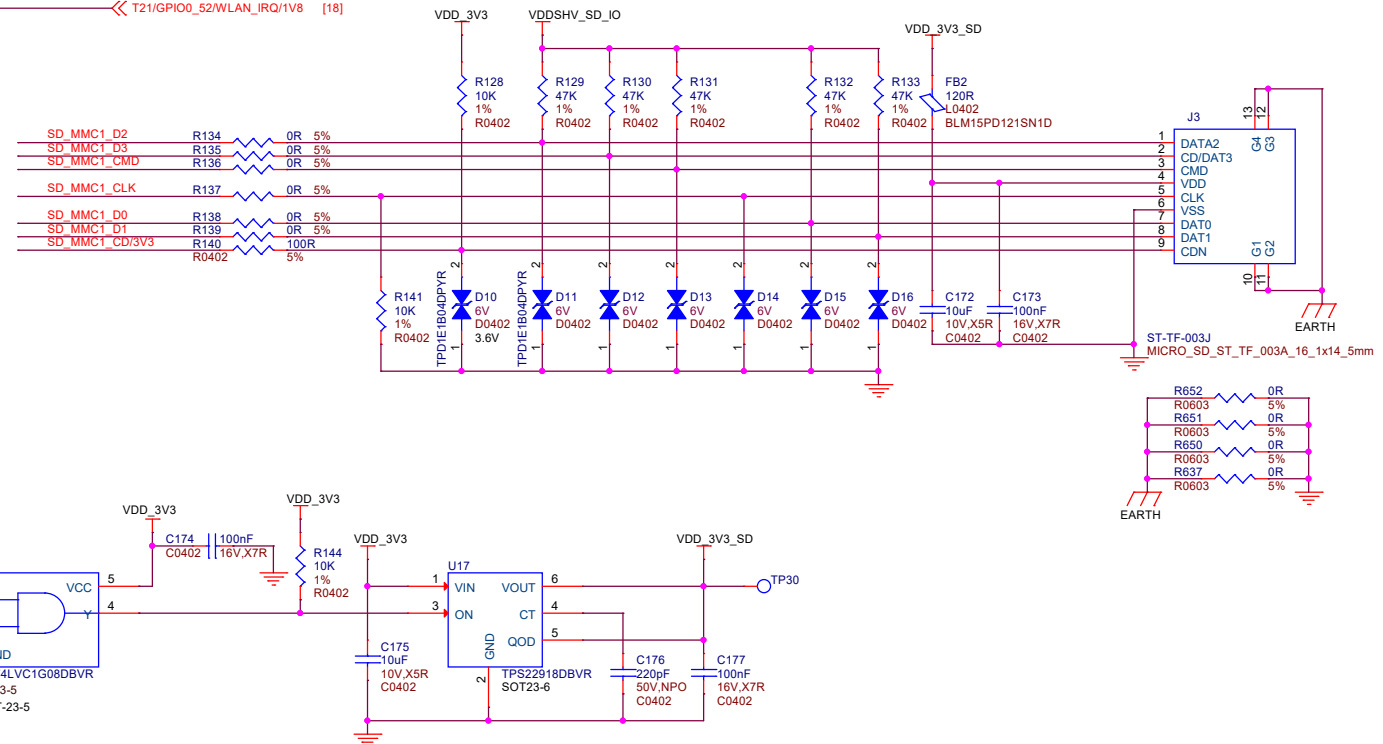
SOC - MMC Interface

U28J



AM62L32BOGHAANBR
BGA373_0d5_12x12x0_87mm
12x12x0_87mm

MICRO SD CARD





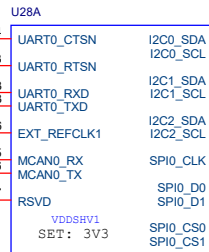
SOC - MCASP and UART



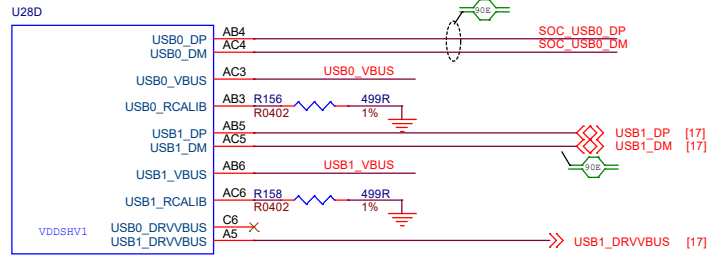
SOC - DSI



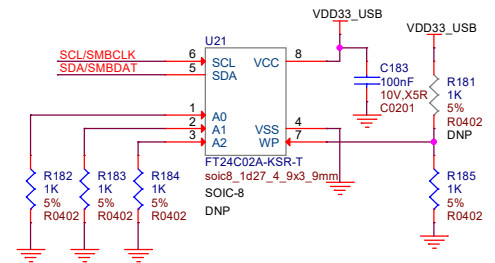
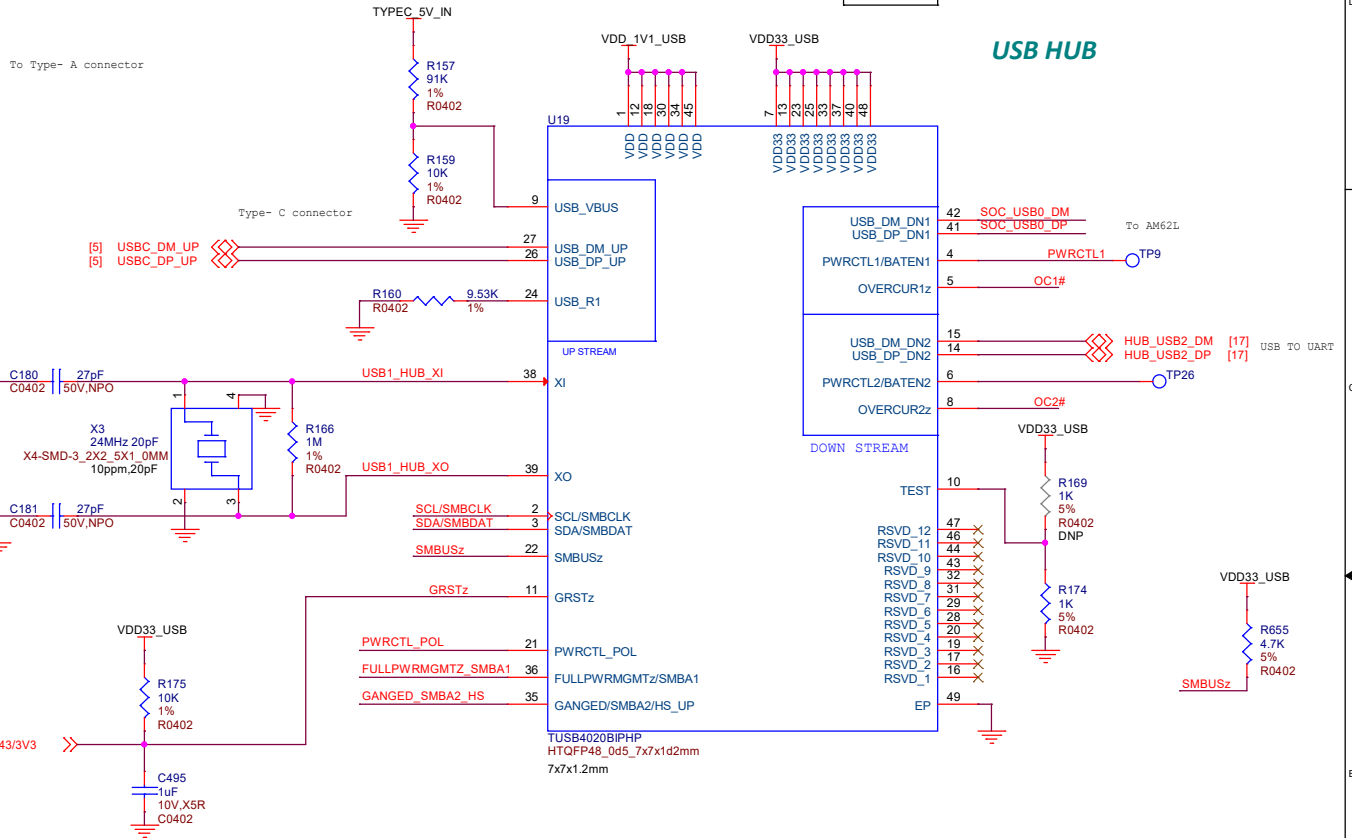
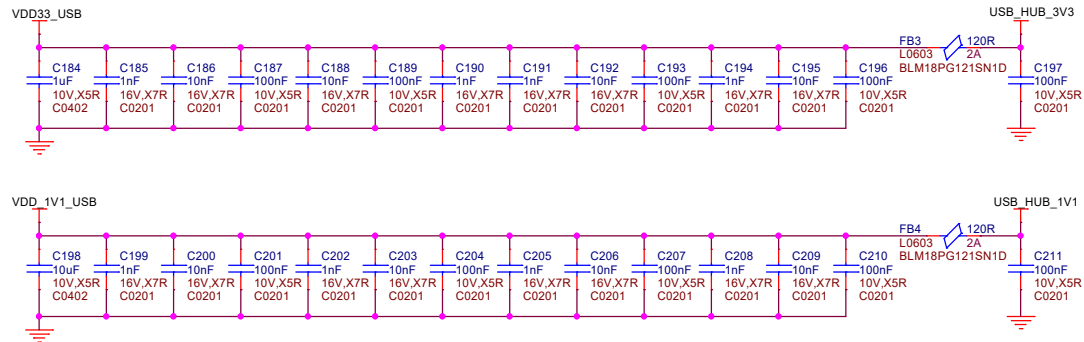
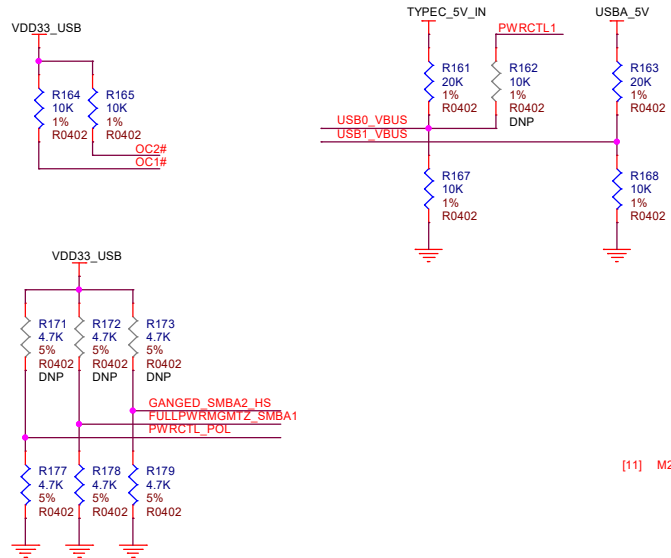
SOC - UART, MCAN , I2C and IOs



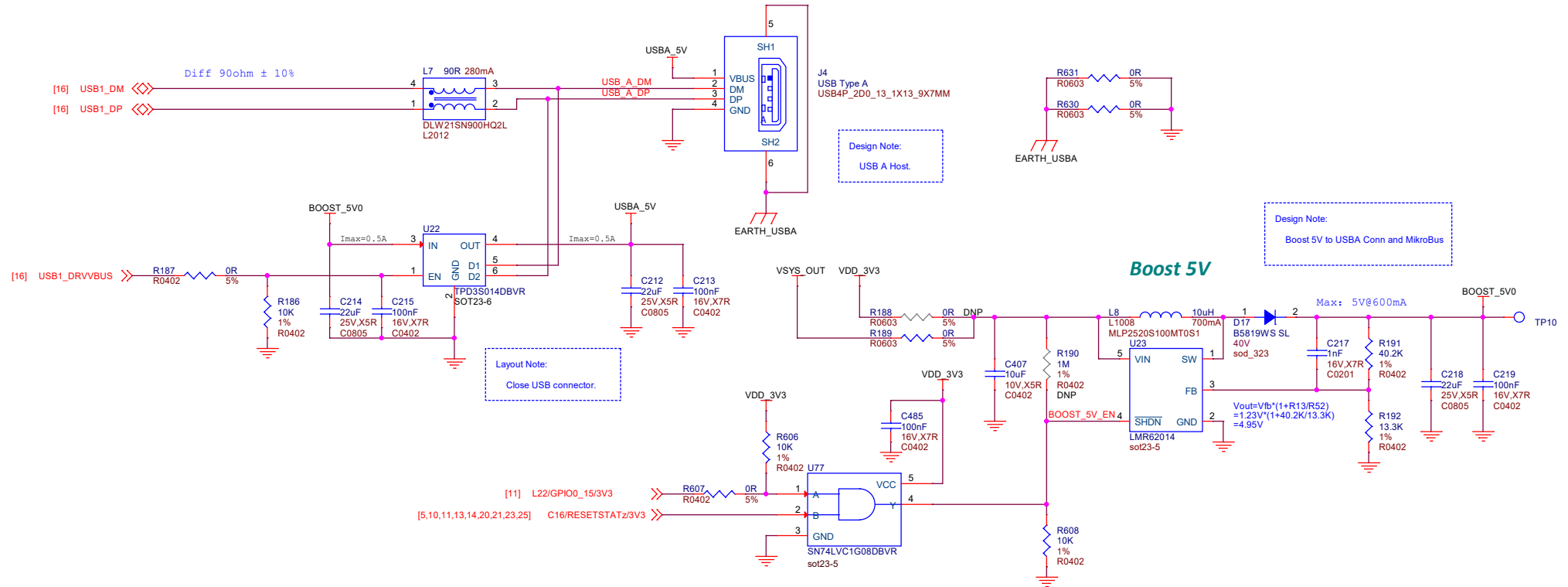
SOC - USB



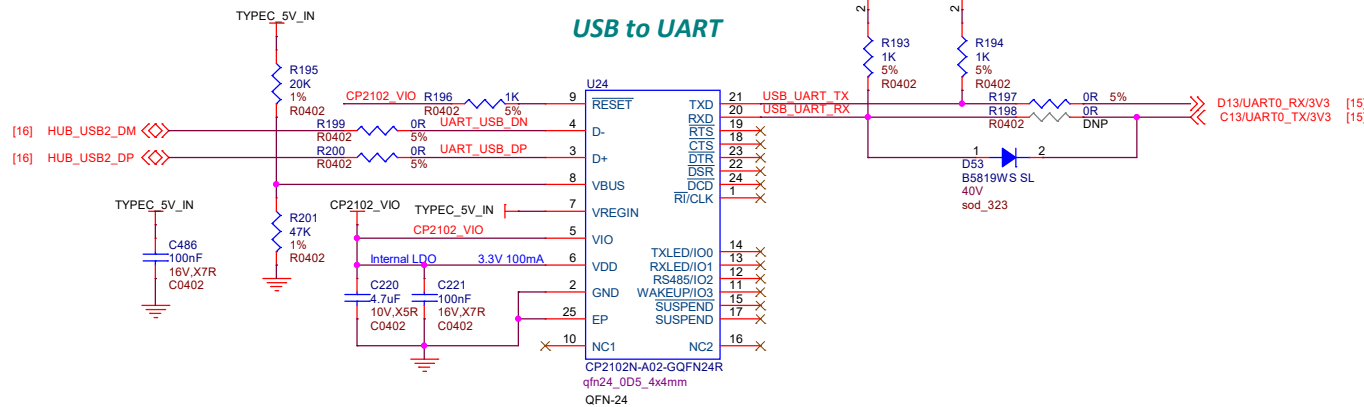
AM62L32BOGHAANBR
BGA373_0d5_12x12x0_87mm
12x12x0_87mm



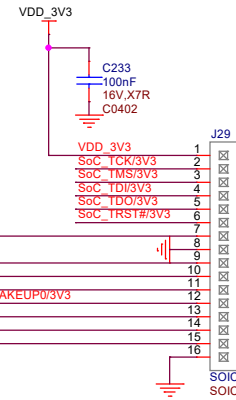
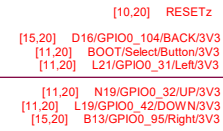
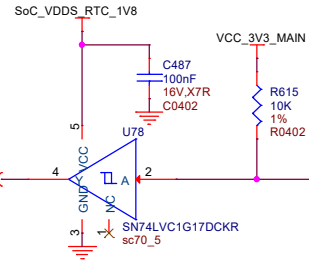
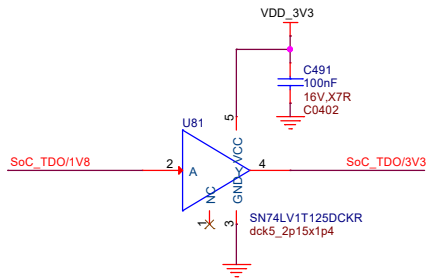
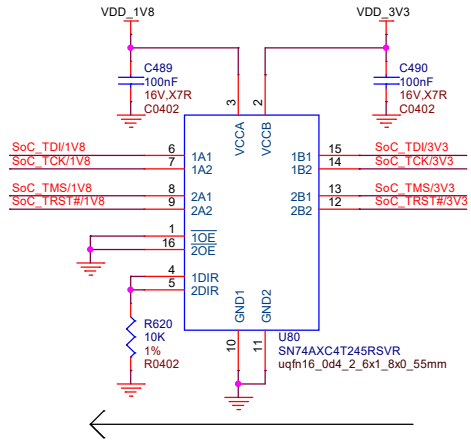
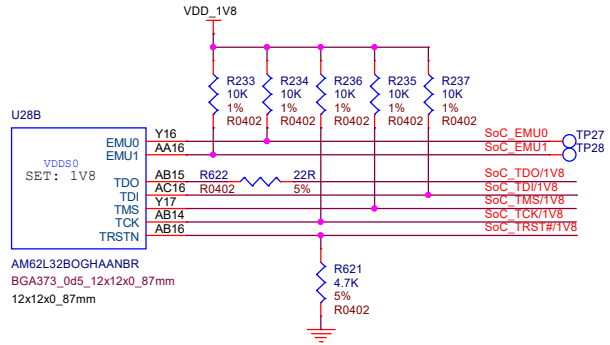
USB TYPEA



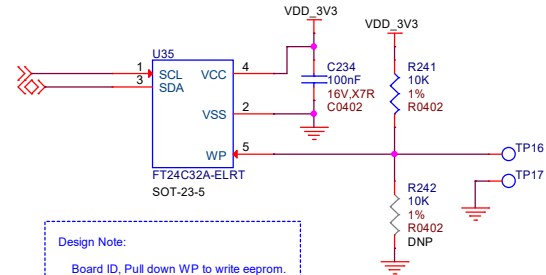
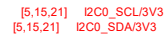
USB to UART



SOC- JTAG



- ```
Pins
1 - VDD
2 - TCK
3 - TMS
4 - TDI
5 - TDO
6 - TRST#
7 - RESETz
8 - GND
9 - button back
10 - button select
11 - button left
12 - EXT_WAKEUP0/3V3
13 - button up
14 - button down
15 - button right
16 - GND
```



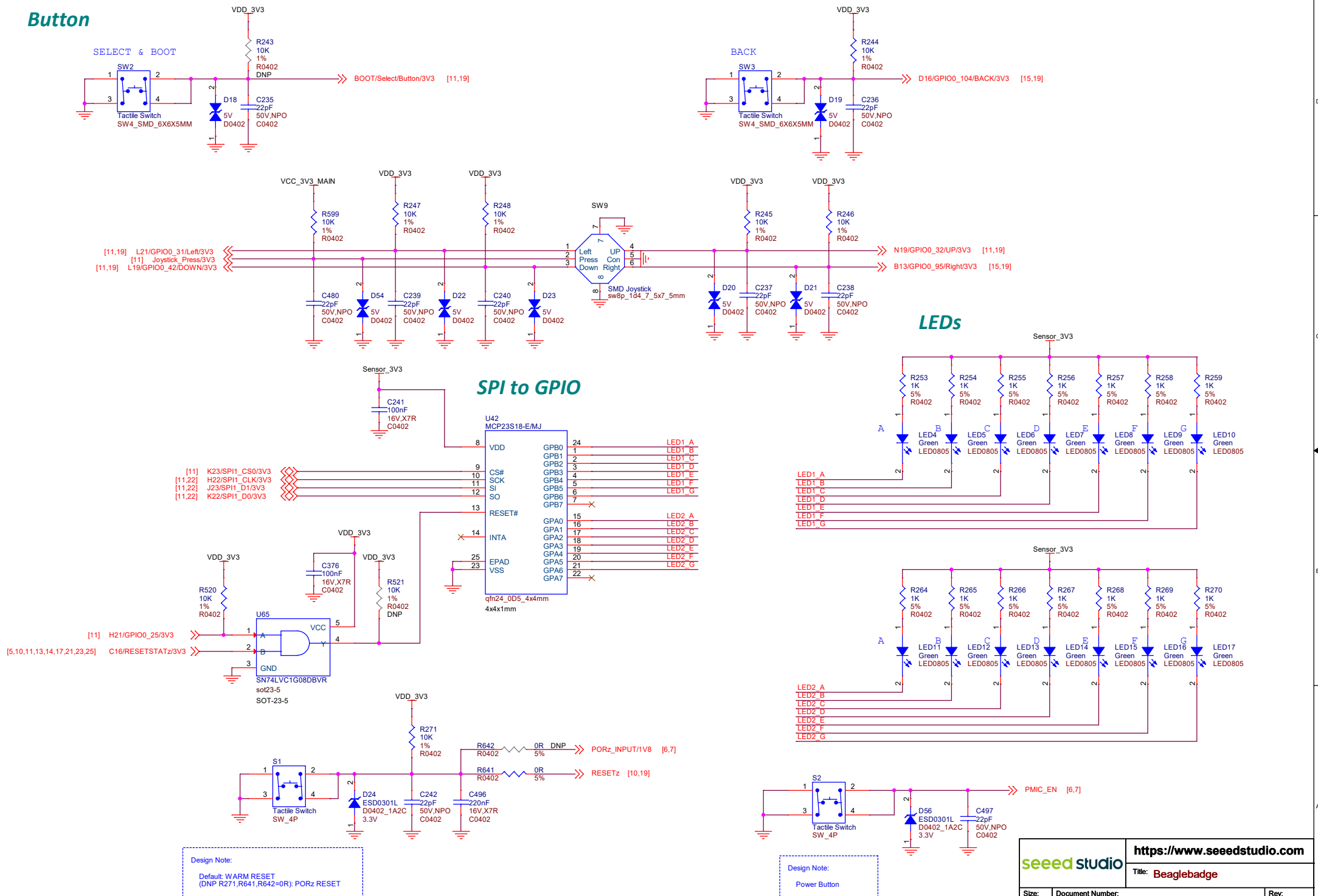
Design Note:  
Board ID, Pull down WP to write eeprom.

## BOARD ID EEPROM 32Kbit



Design Note:  
1V8 Level for JTAG.

## Button



**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

**T & H Sensor**

**Light Sensor**

Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                         |                                                                       |                        |
|--------------------|-------------------------|-----------------------------------------------------------------------|------------------------|
|                    |                         | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                        |
|                    |                         | <b>Title:</b> Beaglebadger                                            |                        |
| <b>Size:</b><br>A3 | <b>Document Number:</b> | 21 Buzzer & IMU & Light                                               | <b>Rev:</b><br>v1.0    |
| <b>Date:</b>       |                         | Tuesday, July 26, 2026                                                | <b>Sheet:</b> 21 of 25 |

**Sensor Power Switch**

VDD\_1V8

R272 10K 1% R0402

U61 VCC

1 2 3

SN74LVC1G08DBVR sot23-5 SOT-23-5

VDD\_1V8

C243 100nF 16V.X7R C0402

VDD\_3V3

R273 10K 1% R0402

C244 10uF 10V.X5R C0402

U32 VIN VOUT 6

ON CT 4

QD 5

GND

TP31

Sensor\_3V3

C245 220pF 50V.NPO C0402

C406 10uF 10V.X5R C0402 DNP

C391 100nF 16V.X7R C0402

**Design Note:**

Vih min = 1V.

**RGB LED**

Sensor\_3V3

R276 1K 5% R0402

R277 1K 5% R0402

R278 2.2K 1% R0402

Q1 2SK3018 SOT-323-3

Q2 2SK3018 SOT-323-3

Q3 2SK3018 SOT-323-3

R282 100K 1% R0402

R283 100K 1% R0402

R284 100K 1% R0402

[15] A11/GPIO0\_83/PWM/3V3

[15] A9/GPIO0\_81/PWM/3V3

[15] B9/GPIO0\_82/PWM/3V3

**Passive Buzzer**

VCC\_3V3\_MAIN VDD\_3V3

R616 0R 5% DNP

R617 0R 5% DNP

VCC\_3V3\_IMU

[5,15,19,21] I2C0\_SCL/3V3

[5,15,19,21] I2C0\_SDA/3V3

R287 0R 5% IMU\_SCL

R288 4.7K 5% R0402

R289 0R 5% IMU\_SDA

VCC\_3V3\_MAIN

R290 0R DNP

R291 0R 5% R0402

R292 0R 5% R0402

R294 0R 5% R0402

VCC\_3V3\_IMU

R295 100K 1% R0402

[22] IMU\_INT/3V3

Software: Default: High INT: low

**IMU Sensor**

VCC\_3V3\_IMU

R288 4.7K 5% R0402

U33 SDO/SA0 14

SCL 13

SDx 12

SCx 11

INT1 9

INT2 10

VDD 8

VDDIO 5

GND1 6

GND2 7

LSM6DS3TR-C LGA14\_0d5\_2\_5x3x0\_86mm

C248 100nF 16V.X7R C0402

**T & H Sensor**

[5,15,19,21] I2C0\_SDA/3V3

[5,15,19,21] I2C0\_SCL/3V3

R326 0R 5% R0402

R327 0R 5% R0402

U40 SDA VDD 3

SCL VSS 4

GND\_EP 5

SHT40 dfn4\_0d8\_1\_5x1\_5x0\_54mm\_rko DNP

C410 100nF 16V.X7R C0402

**Light Sensor**

Sensor\_3V3

C411 100nF 16V.X7R C0402

U45 X2SON4\_0d48\_0\_8x0\_8x0\_4mm

R511 10K 1% R0402

R513 12K 1% R0402

C282 10nF 16V.X7R C0201

TLV9061IDPWR

OUT

+IN

-IN

R573 0R 5% R0402

Sensor\_3V3

C411 100nF 16V.X7R C0402

Q7 ALS-PT19-315C/L177/TR8

SNR2\_1d5\_1\_7x0\_8x0\_6mm

R509 15K 1% R0402

**Design Note:**

Vout(max.) = Vcc - 0.4V

**Layout Note:**

C282 Caps should be close to AM62L.

**Design Note:**

Input: 0 - 2V9 to Output: 0 - 1V6

**seeed studio**

<https://www.seeedstudio.com>

Title: Beaglebadge

Size: A3

Document Number: 21 Buzzer & IMU & Light

Rev: v1.0

Date: Tuesday, July 26, 2026

Sheet: 21 of 25

**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

**T & H Sensor**

**Light Sensor**

Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                  |                                                                       |                     |
|--------------------|------------------|-----------------------------------------------------------------------|---------------------|
|                    |                  | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                     |
|                    |                  | Title: <b>Beaglebadge</b>                                             |                     |
| Size:<br><b>A3</b> | Document Number: | 21 Buzzer & IMU & Light                                               | Rev:<br><b>v1.0</b> |
| Date:              |                  | Tuesday, July 26, 2026                                                | Sheet: 21 of 25     |

**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

**T & H Sensor**

**Light Sensor**

Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                         |                                                                       |                        |
|--------------------|-------------------------|-----------------------------------------------------------------------|------------------------|
|                    |                         | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                        |
|                    |                         | <b>Title:</b> Beaglebadger                                            |                        |
| <b>Size:</b><br>A3 | <b>Document Number:</b> | 21 Buzzer & IMU & Light                                               | <b>Rev:</b><br>v1.0    |
| <b>Date:</b>       |                         | Tuesday, July 26, 2026                                                | <b>Sheet:</b> 21 of 25 |

**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

**T & H Sensor**

**Light Sensor**

Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                         |                                                                       |                        |
|--------------------|-------------------------|-----------------------------------------------------------------------|------------------------|
|                    |                         | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                        |
|                    |                         | <b>Title:</b> Beaglebadger                                            |                        |
| <b>Size:</b><br>A3 | <b>Document Number:</b> | 21 Buzzer & IMU & Light                                               | <b>Rev:</b><br>v1.0    |
| <b>Date:</b>       |                         | Tuesday, July 26, 2026                                                | <b>Sheet:</b> 21 of 25 |

**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

**T & H Sensor**

**Light Sensor**

Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                         |                                                                       |                        |
|--------------------|-------------------------|-----------------------------------------------------------------------|------------------------|
|                    |                         | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                        |
|                    |                         | <b>Title:</b> Beaglebadger                                            |                        |
| <b>Size:</b><br>A3 | <b>Document Number:</b> | 21 Buzzer & IMU & Light                                               | <b>Rev:</b><br>v1.0    |
| <b>Date:</b>       |                         | Tuesday, July 26, 2026                                                | <b>Sheet:</b> 21 of 25 |

**Sensor Power Switch**

Design Note:  
V<sub>IH</sub> min = 1V.

**RGB LED**

Design Note:  
V<sub>out</sub>(max.) = V<sub>cc</sub> - 0.4V

**Passive Buzzer**

**IMU Sensor**

Software:  
Default: High  
INT: low

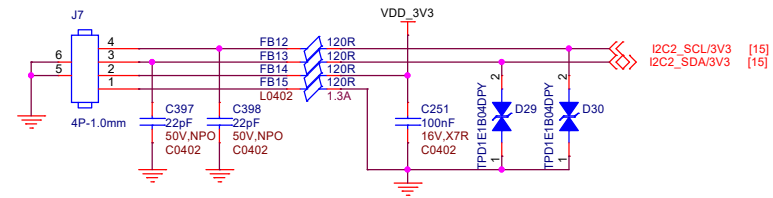
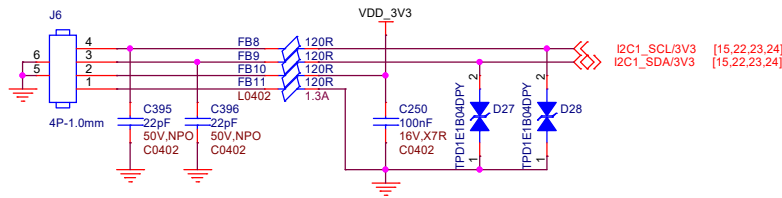
**T & H Sensor**

**Light Sensor**

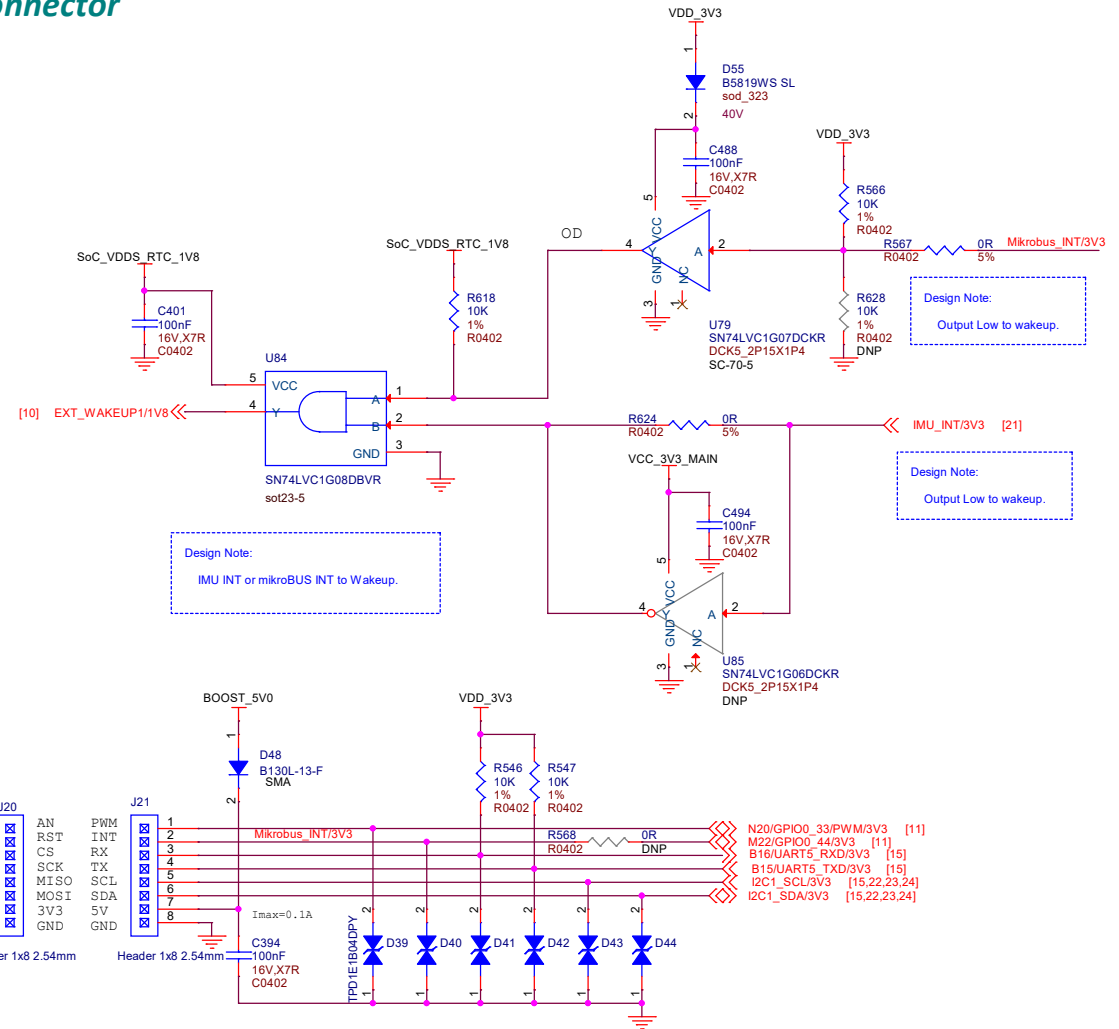
Design Note:  
C282 Caps should be close to AM62L.

Design Note:  
Input: 0 - 2V<sub>9</sub> to Output: 0 - 1V<sub>6</sub>

|                    |                  |                                                                       |                     |
|--------------------|------------------|-----------------------------------------------------------------------|---------------------|
|                    |                  | <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a> |                     |
|                    |                  | Title: <b>Beaglebadge</b>                                             |                     |
| Size:<br><b>A3</b> | Document Number: | 21 Buzzer & IMU & Light                                               | Rev:<br><b>v1.0</b> |
| Date:              |                  | Tuesday, July 26, 2026                                                | Sheet: 21 of 25     |



## QWIIC Connector



## MikroBus

Layout Note:  
Close to epaper connector

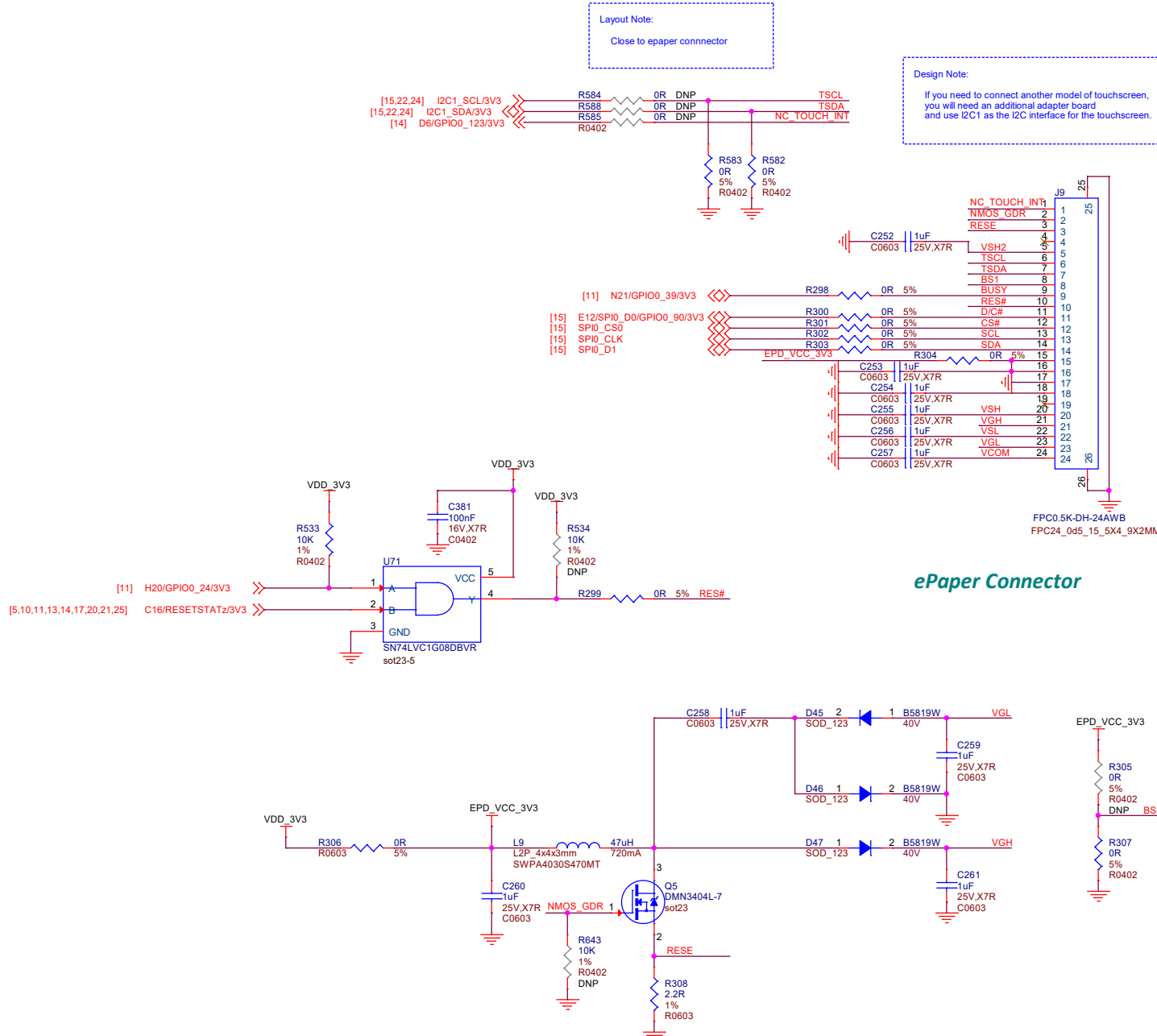
Design Note:  
If you need to connect another model of touchscreen, you will need an additional adapter board and use I2C1 as the I2C interface for the touchscreen.

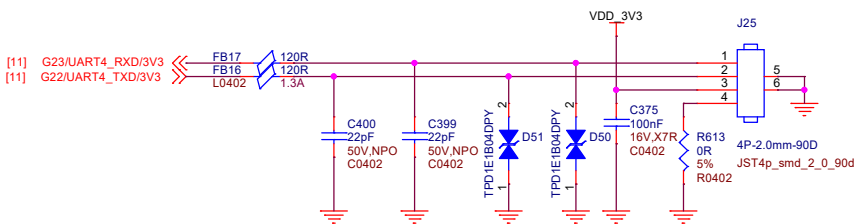
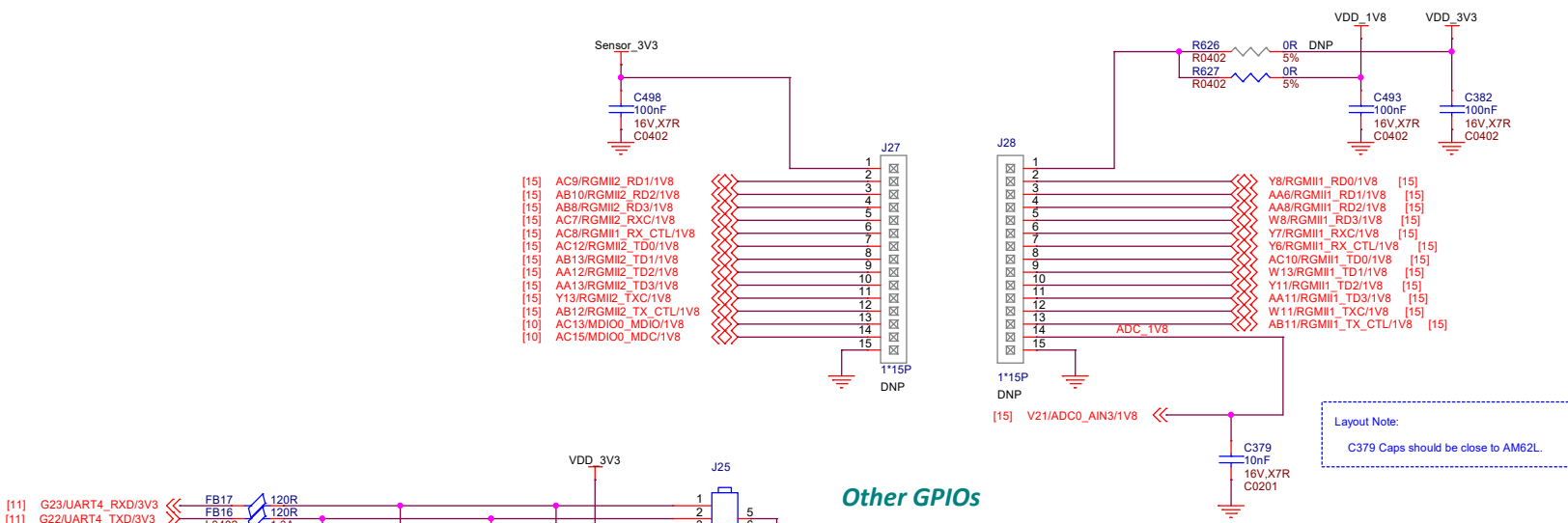
# Input/Output Terminals

| Pin # | Signal | No. |
|-------|--------|-----|
| 1     | NC     |     |
| 2     | GDR    |     |
| 3     | RESE   |     |
| 4     | NC     |     |
| 5     | VSH2   |     |
| 6     | TSCL   |     |
| 7     | TSDA   |     |
| 8     | BS1    |     |
| 9     | BUSY   |     |
| 10    | RES#   |     |
| 11    | D/C#   |     |
| 12    | CS#    |     |
| 13    | SCL    |     |
| 14    | SDA    |     |
| 15    | NC     |     |
| 16    | VDDIO  |     |
| 17    | VCI    |     |
| 18    | VSS    |     |
| 19    | VDD    |     |
| 20    | VPP    |     |
| 21    | VSH1   |     |
| 22    | VSL    |     |
| 23    | VGL    |     |
| 24    | VCOM   |     |

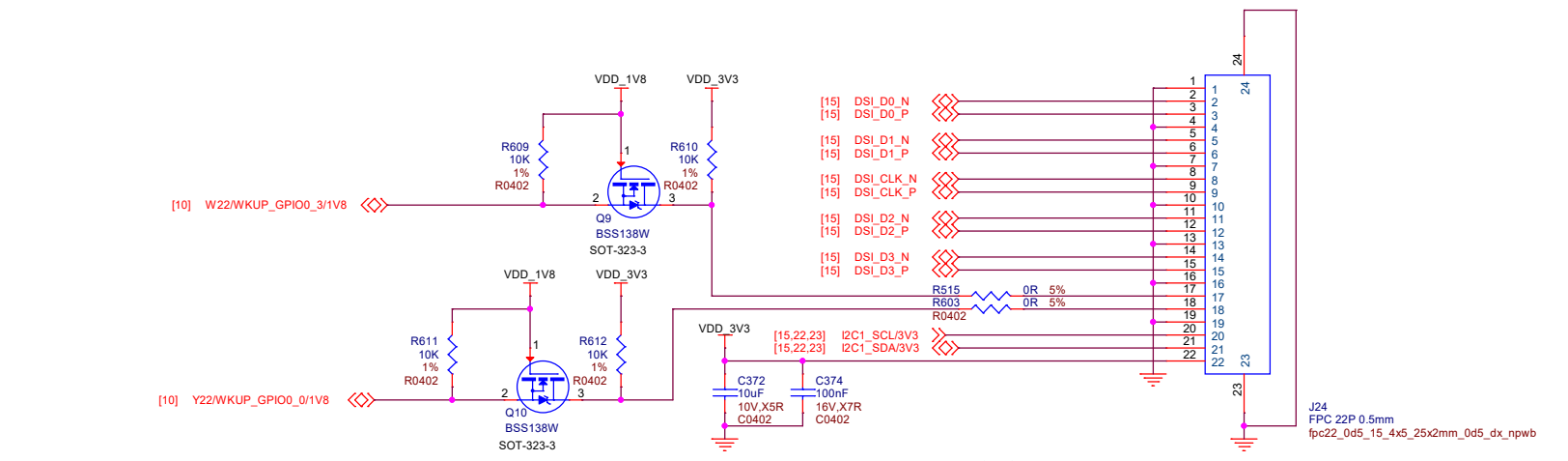
## ePaper Connector

Design Note:  
BS1 = 0, 4-Line SPI Mode  
BS1 = 1, 3-Line SPI Mode





### GROVE Connector



### DSI Connector



